Lab Instructions: Installation and Validation of JDK to Execute First Java Code

Duration: 90 minutes
Objective:

- Install and validate the Java Development Kit (JDK) on your system.
- Write, compile, and execute your first Java program using a text editor.
- Begin mastering basic Java syntax through a guided exercise.

Part 1: Installation and Validation of JDK (30 minutes)

Step 1: Download and Install JDK

1. **Download JDK:**
    - Visit the official Oracle website: Oracle JDK Downloads.
        1. https://www.oracle.com/java/technologies/downloads
    - Download the latest version of the JDK suitable for your operating system (Windows, macOS, or Linux).
2. **Install JDK:**
    - Run the installer and follow the on-screen instructions.
    - Take note of the installation directory (e.g., `C:\Program Files\Java\jdk-<version>` on Windows).

Step 2: Set Up Environment Variables

1. **For Windows:**
    - Open the Start menu, search for "Environment Variables," and click "Edit the system environment variables."
    - In the System Properties window, click "Environment Variables."
    - Under "System variables," find the `Path` variable and click "Edit."
    - Add the path to the JDK's `bin` directory (e.g., `C:\Program Files\Java\jdk-<version>\bin`).
    - Click "OK" to save changes.

Step 3: Validate JDK Installation

1. Open a terminal or command prompt.
2. Run the following command to check the Java version:
   `java -version`
    - You should see the installed JDK version displayed.
3. Run the following command to check the Java compiler:
4. `javac -version`
    - You should see the version of the Java compiler.

Part 2: Writing and Executing Your First Java Program (30 minutes)

Step 1: Write Your First Java Program

1. Open a text editor (e.g., Notepad on Windows, TextEdit on macOS, or any code editor like VS Code).
2. Type the following code:

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

3. Save the file as `HelloWorld.java` in a directory of your choice (e.g., `C:\JavaProjects`).

Step 2: Compile and Run the Program

1. Open a terminal or command prompt.
2. Navigate to the directory where you saved `HelloWorld.java`:
   `cd C:\JavaProjects`
3. Compile the Java program:
   `javac HelloWorld.java`
   - This will generate a `HelloWorld.class` file.
4. Run the compiled program:
   `java HelloWorld`
   - You should see the output: `Hello, World!`

Part 3: Mastering Basic Java Syntax

1. **Create a New Java Program:**
   - Open your text editor and create a new file named `BasicSyntax.java`.
   - Write the following code to practice basic Java syntax:

```
public class BasicSyntax {
    public static void main(String[] args) {
        // Declare and initialize variables
        int number = 10;
        double decimal = 5.5;
        String text = "Java is fun!";
        // Print variables
        System.out.println("Number: " + number);
        System.out.println("Decimal: " + decimal);
        System.out.println("Text: " + text);
        // Perform arithmetic operations
        int sum = number + (int) decimal; // Casting double to int
        System.out.println("Sum: " + sum);
        // Use a loop
        for (int i = 0; i < 5; i++) {
            System.out.println("Loop iteration: " + i);
        }
        // Use a conditional statement
        if (number > 5) {
            System.out.println("Number is greater than 5");
        } else {
            System.out.println("Number is 5 or less");
        }
```

```
    }}
```

2. **Compile and Run the Program:**
   - Save the file and compile it:

     ```
     javac BasicSyntax.java
     ```

   - Run the program:

     ```
     java BasicSyntax
     ```

   - Observe the output and ensure you understand each part of the code.
3. **Modify the Code:**
   - Add a new variable of type `boolean` and use it in a conditional statement.